

PHYS 225

Week 16, Lecture 14:
Maxwell's Equations and Special Relativity



UNIVERSITY OF
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URBANA-CHAMPAIGN

Reminders

- **Last Homework** 13 due tomorrow 5/6 (note Wed. due date!)
 - Office Hours on Wednesday Loomis 258.
Office Hours on Thursday Loomis 147.
 - **Study sessions** are scheduled on **Friday, May 8, at your usual Discussion times in Loomis 139.**
 - **Final exam:** Tuesday, May 12, 1:30 - 4:30 pm,
D5 A, B, E, J – Noyes Lab 217.
D5 F, K, M, N, R, S – Noyes Lab 100.
- 
- An aerial satellite map of a campus area. A blue path with circular markers is overlaid on the map, starting from the bottom right and moving towards the top left. Various buildings, parking lots, and green spaces are visible. Labels for streets like 'W Green St' and 'W Orange St' are present. There are also small icons and text labels for specific locations, some of which are highlighted with colored circles.



- **Bonus assignment** due in Gradescope by 11:59 pm on 5/11
- Feedback on the Learning Experience (FLEX) survey open: please fill it out, it helps us make this course better!

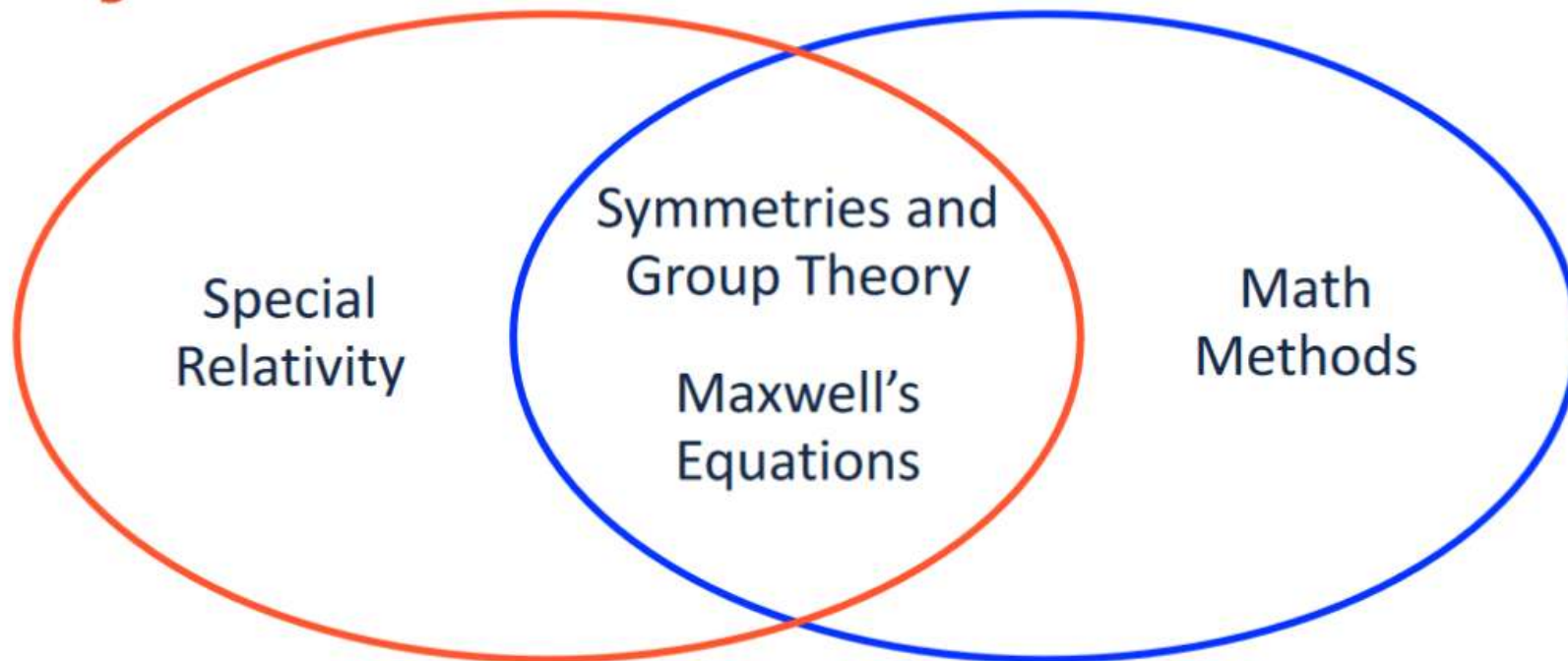
FLEX surveys

- I will now leave the room for ~15 minutes so that you can fill out your FLEX surveys!
- Go to go.illinois.edu/flex
- All responses are completely anonymous.
- We care about your feedback and respect your voice in shaping the course. We need to know what worked and what didn't to make future classes even better!
- If you have the energy to fill out the free-response questions, those are the most helpful.

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Maxwell's Equations and Special Relativity

Objectives:



One Minute Paper: name one topic from each category that you learned about in this course for the first time

Maxwell's equations in differential form

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

Gauss's Law

$$\nabla \cdot \mathbf{B} = 0$$

Gauss's Law for Magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's Law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Ampere's Law (to derive in discussion)

We are almost to our goal of combining E+M with relativity!

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Maxwell's Equations in vacuum

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

The plan: use vector calculus identities to decouple these equations (find separate equations for $\mathbf{E}$ and $\mathbf{B}$)

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Maxwell's Equations in vacuum

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Take curl of both sides:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (\text{HW 12})$$

$$\nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) \quad (\text{exchange partials})$$

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Maxwell's Equations in vacuum

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Take curl of both sides:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

$$\nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

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Maxwell's Equations in vacuum

$$\nabla \cdot \mathbf{E} = \cancel{\frac{\rho}{\epsilon_0}} \mathbf{0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \cancel{\mu_0 \mathbf{J}} \mathbf{0} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Take curl of both sides:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\cancel{\nabla \cdot \mathbf{E}}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

$$\nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

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The wave equation

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\Rightarrow \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0 \quad \text{the speed of light, } c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$\Rightarrow \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0$$

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The wave equation

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\Rightarrow \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0$$

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = 0$$

← identical manipulations give

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Invariance of the wave equation

In natural units,

$$\begin{aligned}\frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} &= 0 \\ \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} &= 0\end{aligned}$$

Solutions are electromagnetic waves that propagate at speed $c = 1$...
but in which frame?

Define the 4-gradient operator $\partial_\mu \equiv (\partial_t, \partial_x, \partial_y, \partial_z)$

Transforms like a 4-vector just like ordinary gradient transforms like
a 3-vector (Discussion 9)

Define the d'Alembertian $\partial_\mu \partial^\mu \equiv g^{\mu\nu} \partial_\mu \partial_\nu = \partial_t^2 - \nabla^2$ minus sign from the metric!

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Invariance of the wave equation

In natural units,

$$\begin{aligned}\partial_\mu \partial^\mu \mathbf{E} &= 0 \\ \partial_\mu \partial^\mu \mathbf{B} &= 0\end{aligned}$$

Just like the Laplacian is invariant under rotations (HW 9 and 12), the d'Alembertian is invariant under Lorentz transformations.

A change of inertial frames will mix $\mathbf{E}$ and $\mathbf{B}$ in linear combinations (Discussion 12), but the wave equation is invariant:

The speed of electromagnetic waves is the same in every inertial frame!

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Solving the wave equation

Let's try our new trick of Fourier-transforming (Lecture 13):

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 e^{-ik_\mu x^\mu}$$

amplitude (constant) 4-wavevector (Discussion 13)

$$\partial_\nu (e^{-ik_\mu x^\mu}) = \delta^\mu_\nu (-ik_\mu) (e^{-ik \cdot x}) = -ik_\nu e^{-ik \cdot x} \quad (\text{HW 13})$$

$$\partial^\nu \partial_\nu e^{-ik \cdot x} = (-i)^2 k^\nu k_\nu e^{-ik \cdot x} = -k^2 e^{-ik \cdot x}$$

$$\partial^\nu \partial_\nu \mathbf{E} = 0 \implies k^2 = 0$$

light is described by a massless 4-vector: photons!!

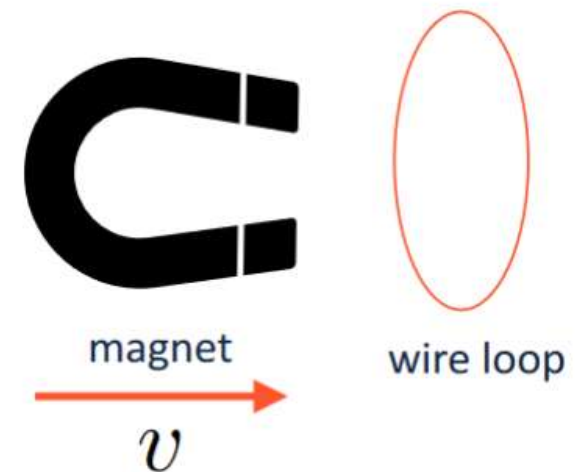
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Einstein has the last word

ON THE ELECTRODYNAMICS OF MOVING BODIES

BY A. EINSTEIN

June 30, 1905



Take, for example, the reciprocal electrodynamic action of a magnet and a conductor...

if the magnet is in motion and the conductor at rest, there arises an electric field producing a current. But if the magnet is stationary and the conductor in motion, no electric field arises in the neighborhood of the magnet. We find an electromotive force, which gives rise to electric currents of the same path and intensity as those produced by the electric forces in the former case.

We see that electromotive force plays in the developed theory merely the part of an auxiliary concept, which owes its introduction to the circumstance that electric and magnetic forces do not exist independently of the state of motion of the system of co-ordinates. Furthermore it is clear that the asymmetry mentioned in the introduction as arising when we consider the currents produced by the relative motion of a magnet and a conductor, now disappears.

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Maxwell's Equations and Special Relativity

Summary

- Maxwell's equations in vacuum imply a wave equation where **E** and **B** travel at c
- The wave equation is relativistically invariant: speed c in all frames!
- The solutions to the wave equation are massless 4-vectors which describe photons
- Electrodynamics is intrinsically relativistic!
- It's been a pleasure teaching this class. Best of luck on the final, and see you around Loomis!

One Minute Paper: what was your favorite topic covered in this class?

Final exam tips

- Write **legibly**: any scratched-out or crossed-out material will be ignored and not graded.
- **Explain your work!** If the graders have to guess what you mean by an equation, you will not receive full credit.
- **Define your variables and symbols.** If you introduce a new symbol (like the Lorentz factor γ , or a time coordinate t) that is not given in the problem statement, you need to explain what it means.
- No problem requires pages of algebra. Some problems may only require interpreting a single formula. The length of the midterm solutions is a good guide.
- There will be an opportunity for bonus points. The bonus problems are not necessarily harder! If you know the material, definitely attempt them.